

ARTEMIS: Adaptive Bitrate Ladder Optimization for Live Video Streaming

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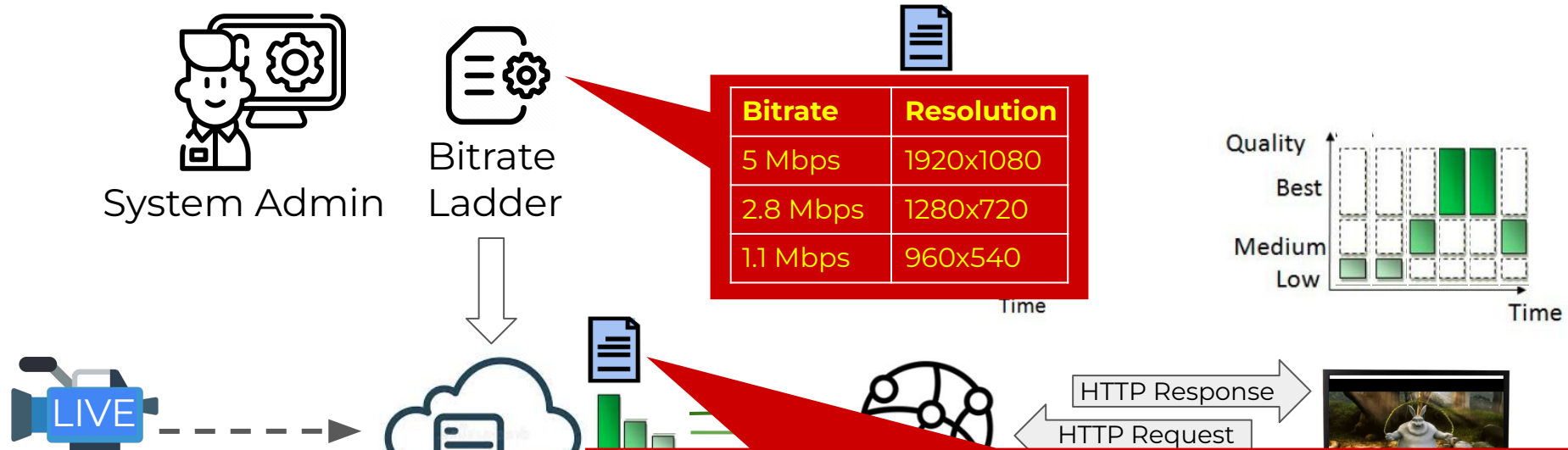
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Live Video Streaming Pipeline

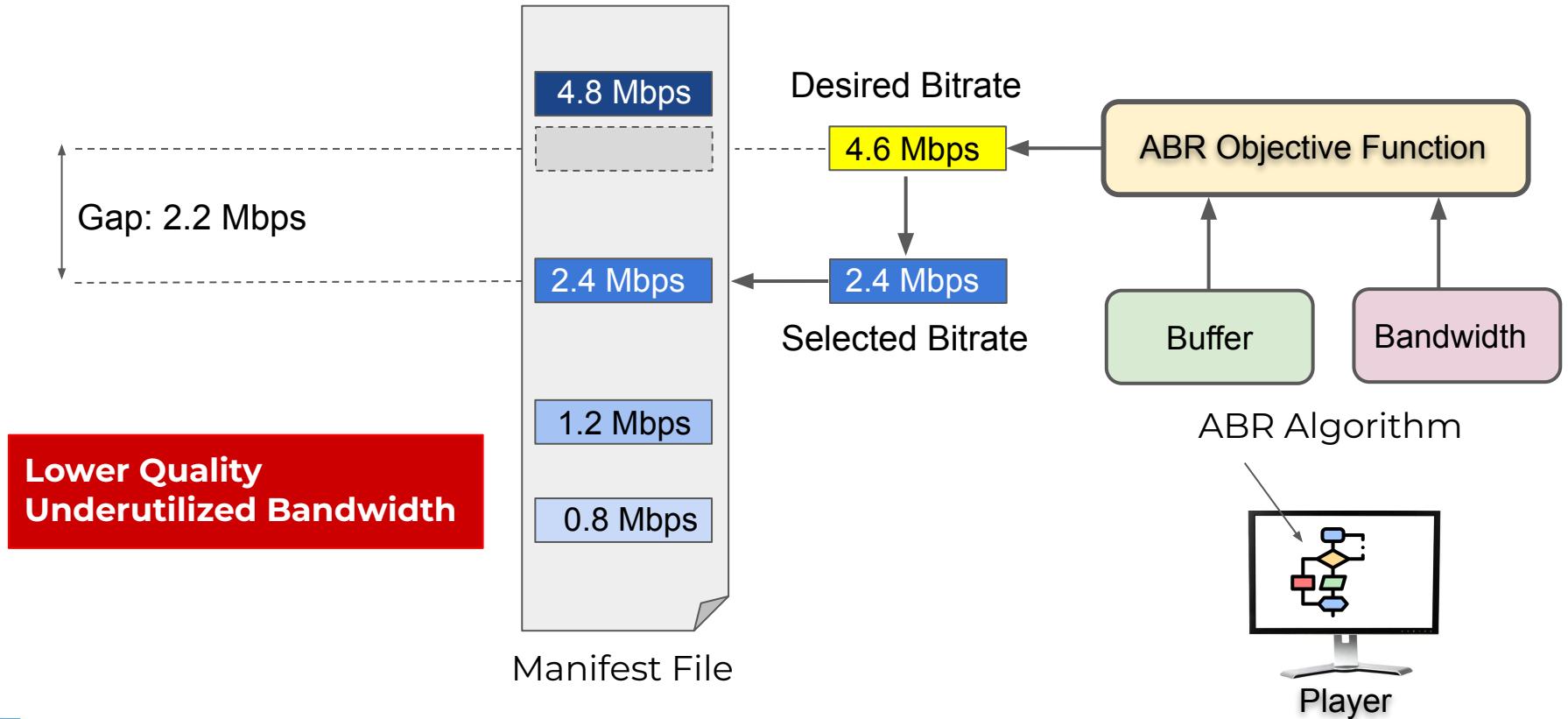


- Fixed bitrate ladder

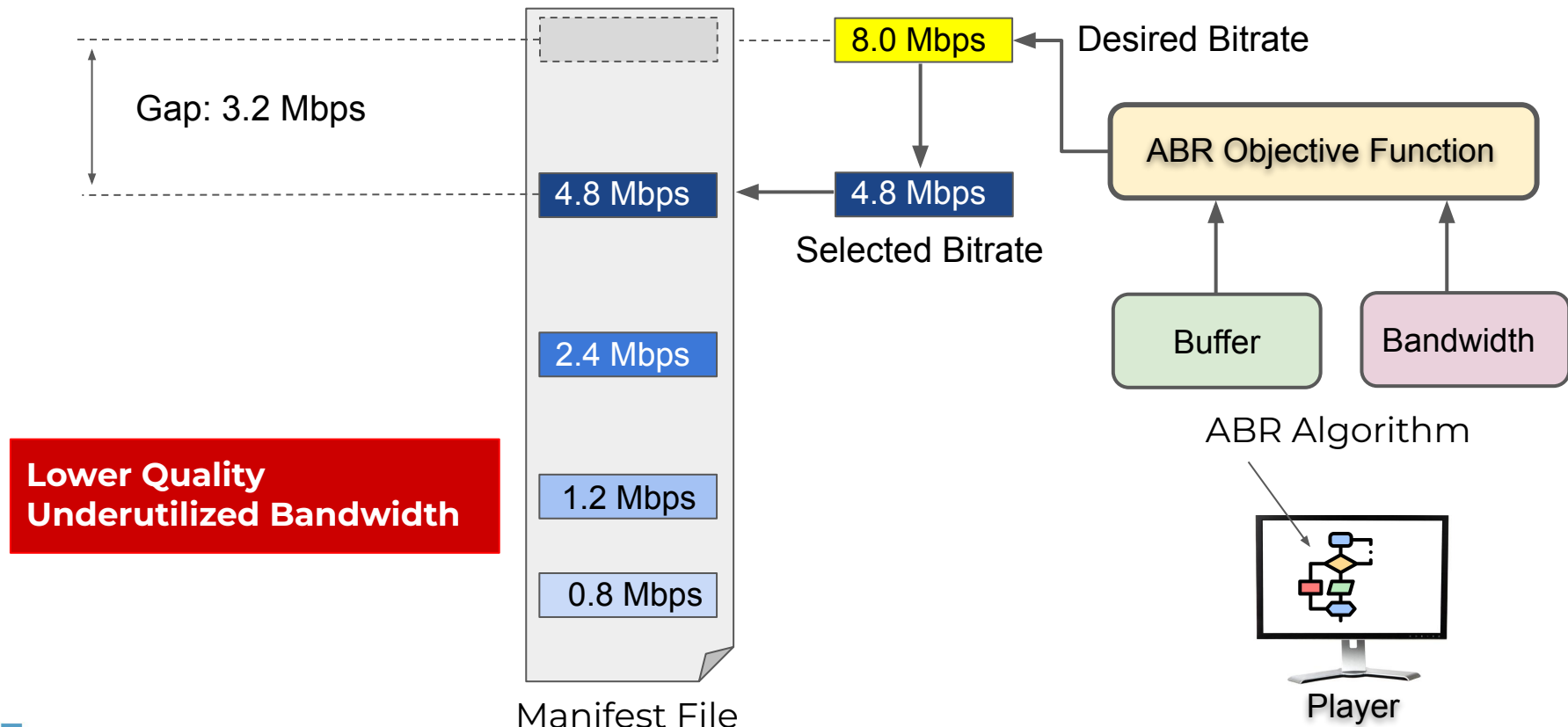
- Content-aware, e.g., Netflix
- Context-aware, e.g., [Lebrecht et al.]
- Agnostic of the content at all

Manifest includes representations: different versions of the content optimized for various playback conditions, e.g., **different bitrates** and **resolutions**

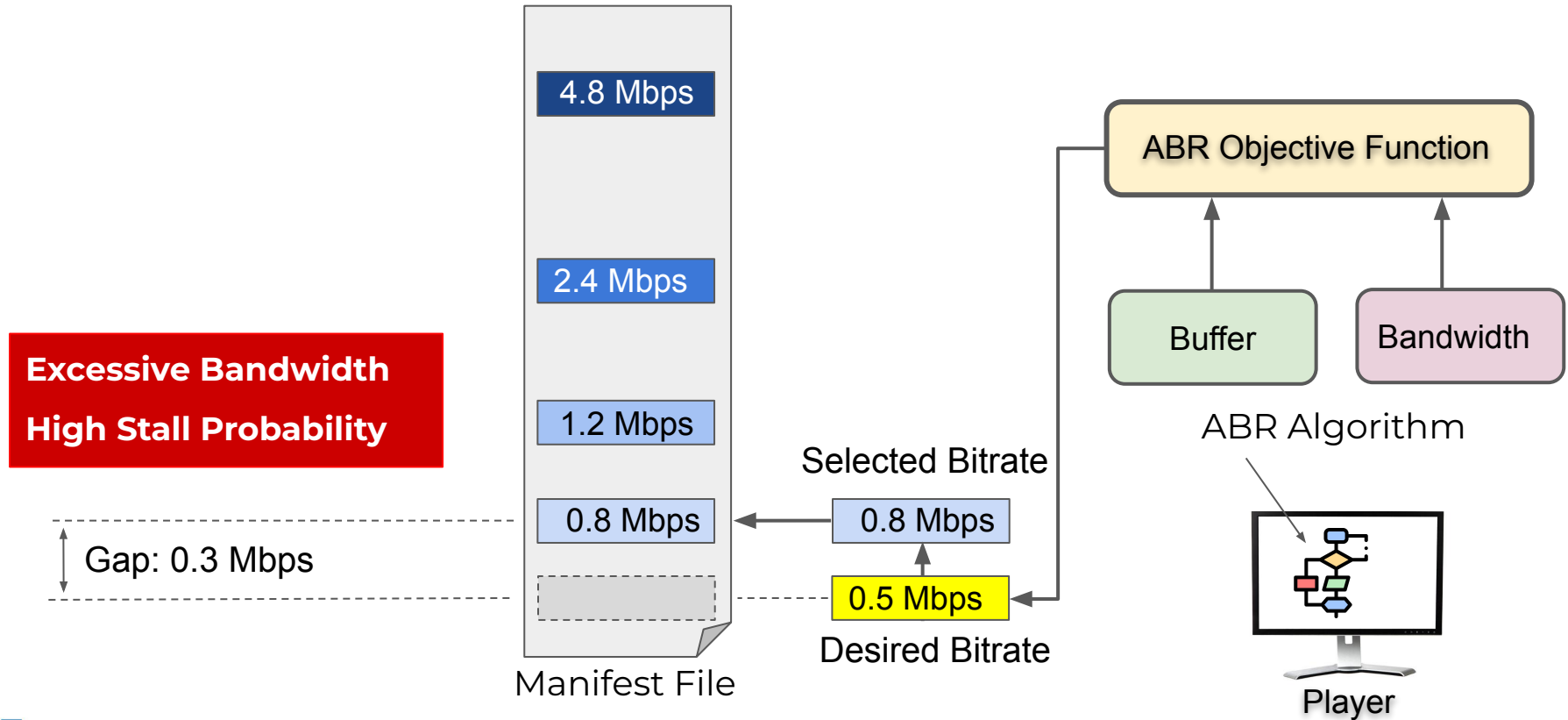
Bitrate Selection by an Adaptive Bitrate (ABR) Algorithm



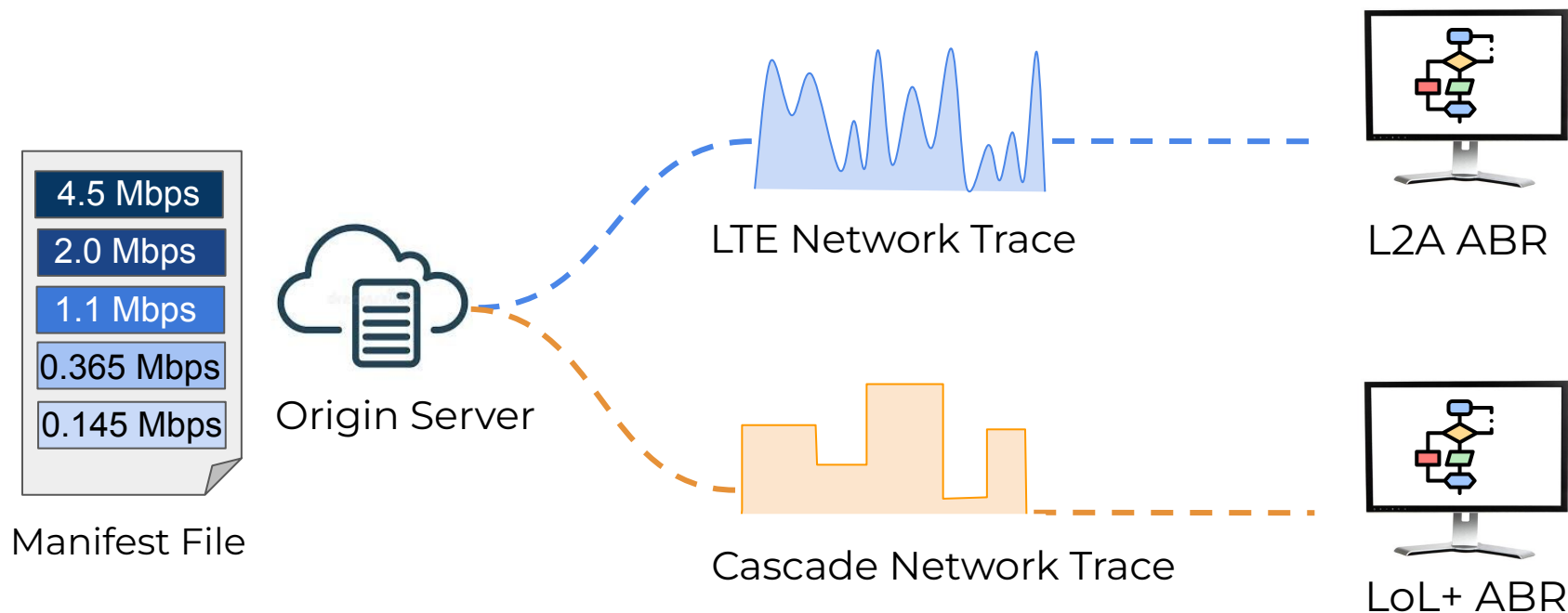
The Highest Bitrate Might be Too Low

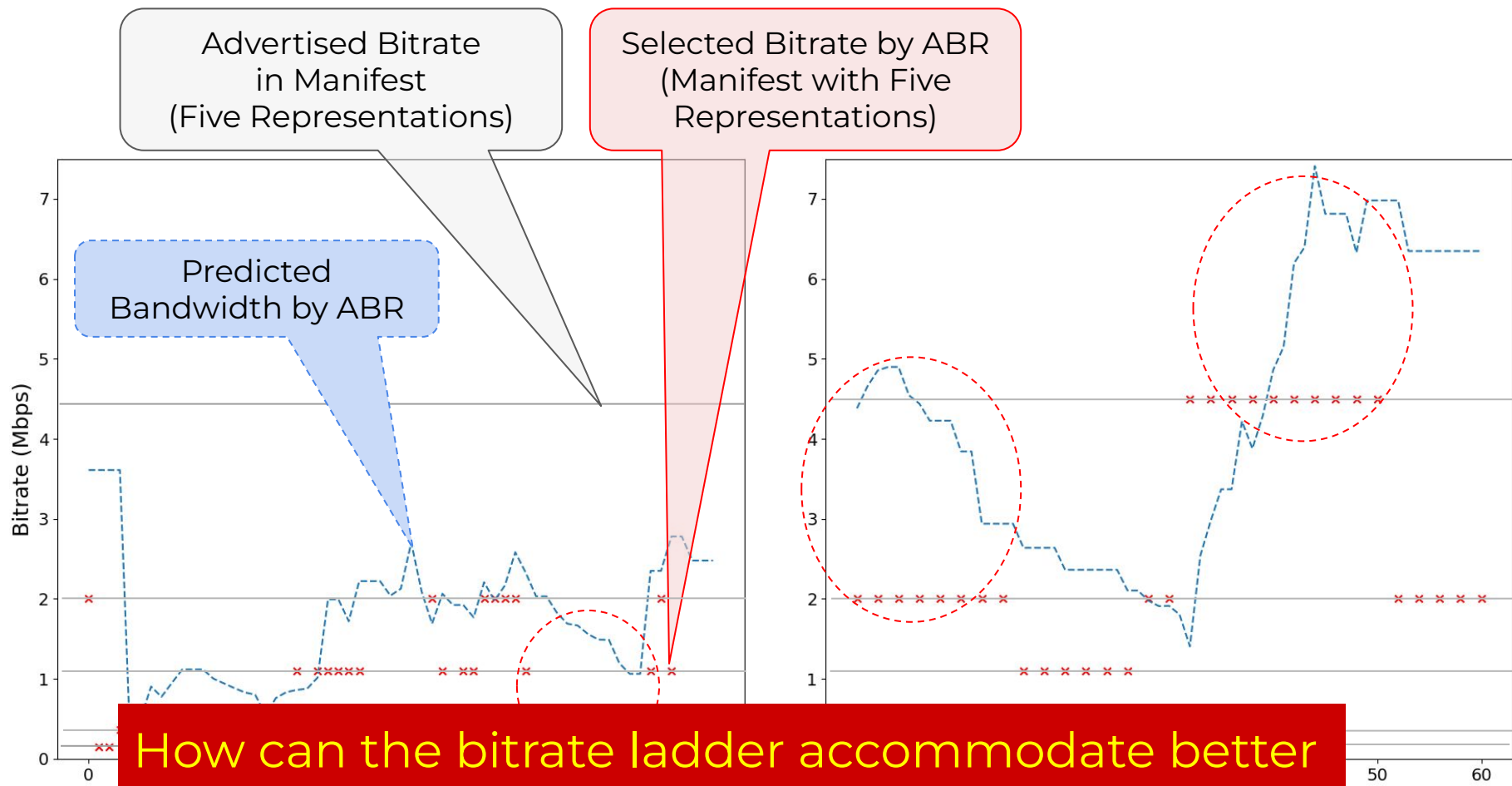


The Lowest Bitrate Might be Too High

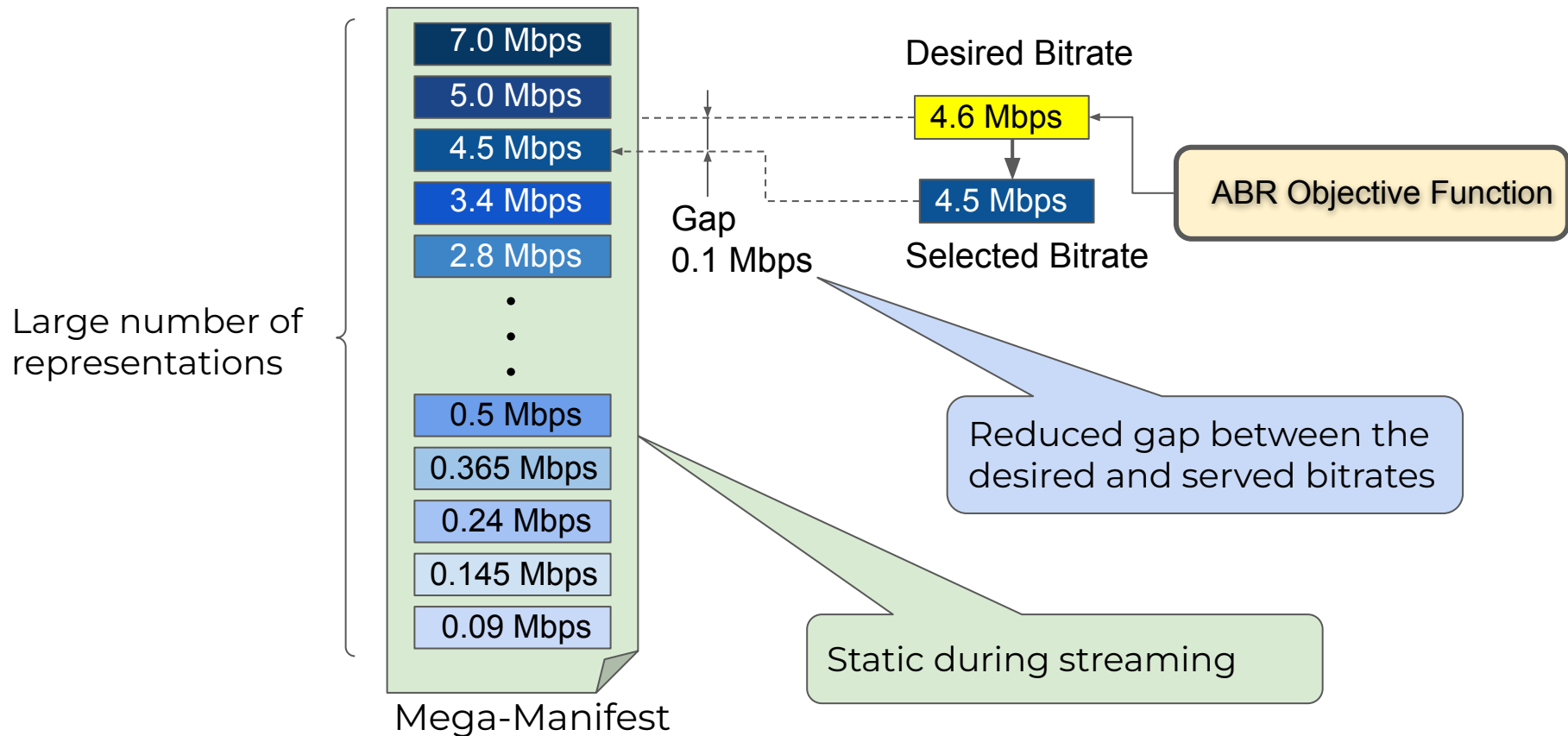


Advertising Five Representations by the Manifest

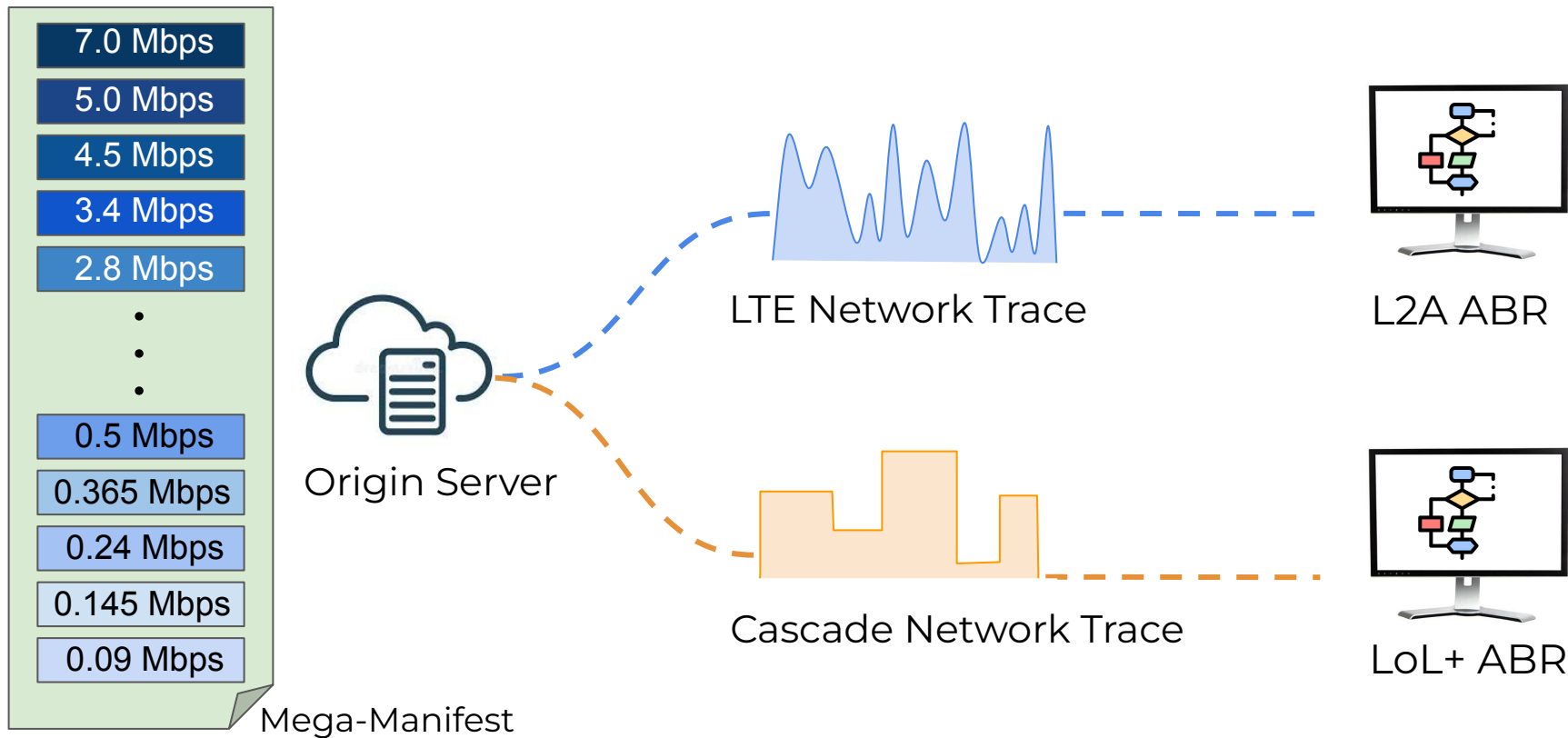




Mega-Manifest for Better Bitrate Alignment



Advertising 19 Representations by the Mega-Manifest

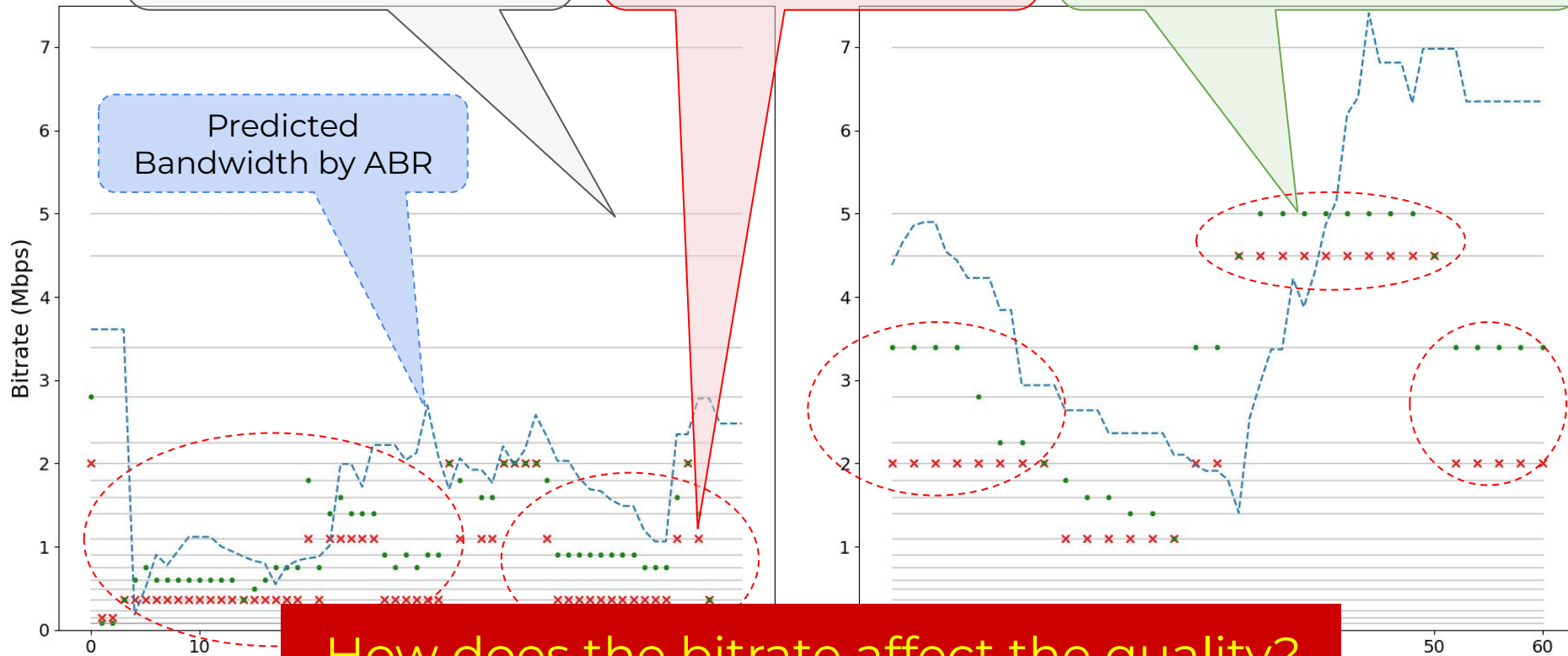


Advertised Bitrate
in Mega- Manifest
(19 Representations)

Selected Bitrate by ABR
(Manifest with Five
Representations)

Selected Bitrate by ABR
(Mega-Manifest with 19
Representations)

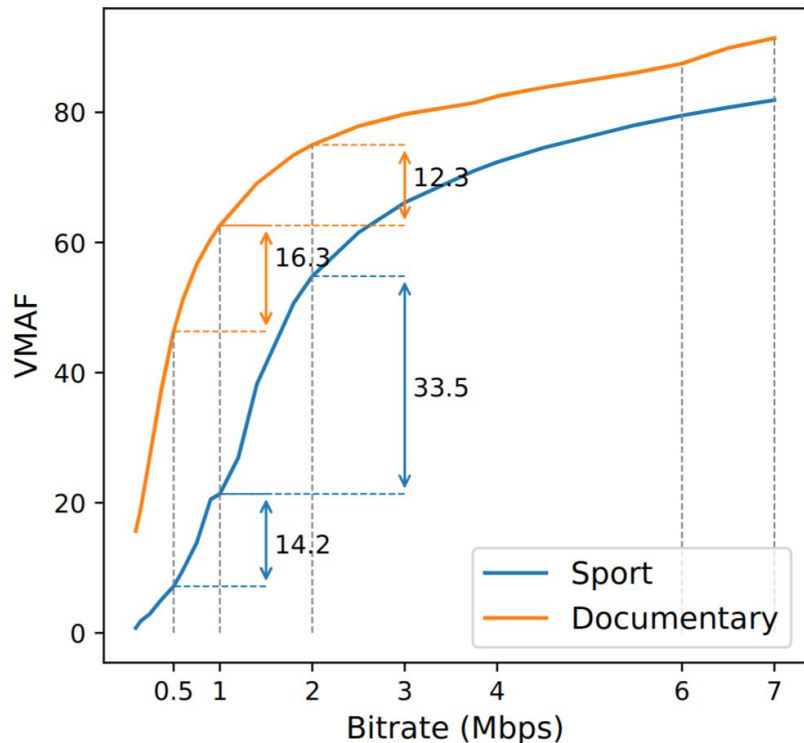
Predicted
Bandwidth by ABR



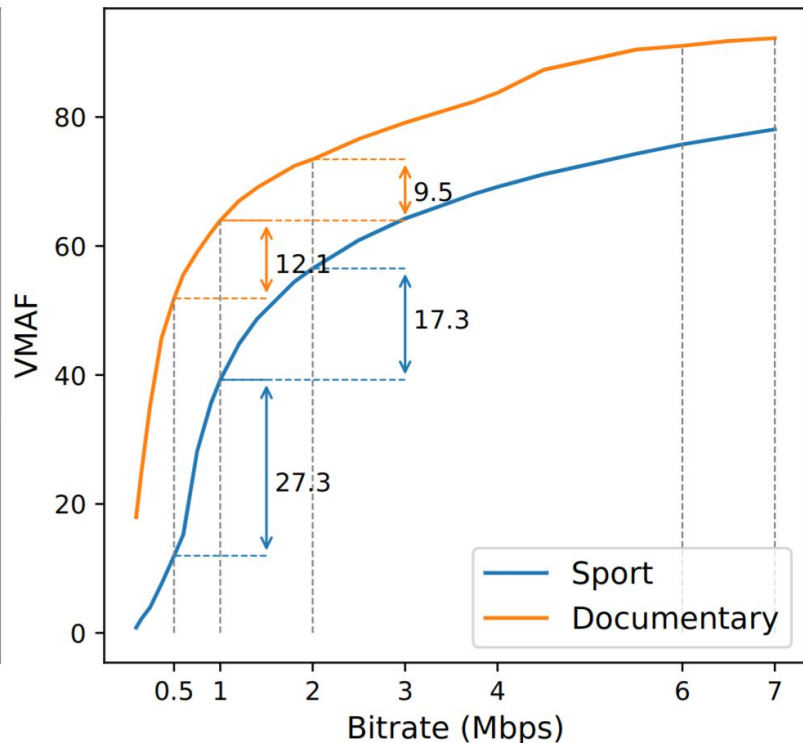
How does the bitrate affect the quality?

Dependence of Video Quality on the Bitrate

VMAF: Video Multimethod Assessment Fusion



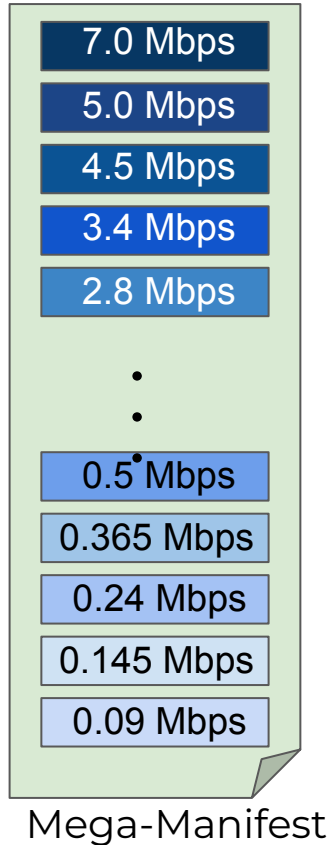
(a) Resolution 1920x1080



(b) Resolution 1280x720

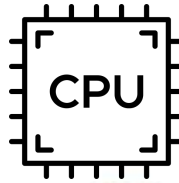
A good bitrate ladder should be aware of video quality

Impact of the Mega-Manifest on the Encoder



Encoding the content into all representations of the mega-manifest in live streaming **is time-consuming**

High Computation



Origin Server
(Encoding & Packaging)

High Bandwidth



High Storage

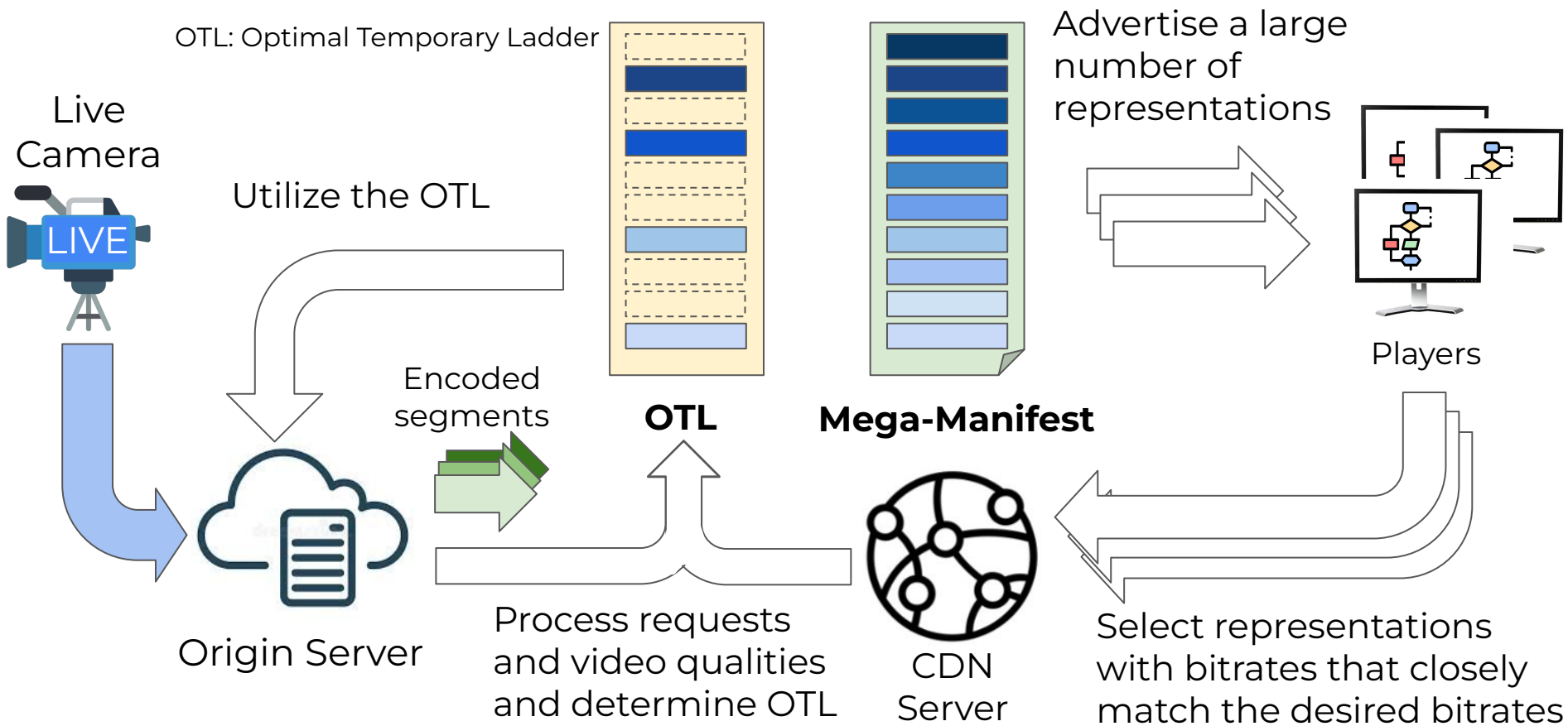


CDN Server

ARTEMIS Aims To:

- Adaptively select an **optimal subset of the mega-manifest representations** by using:
 - Predicted **quality indicator** as the peak signal-to-noise ratio (PSNR)
 - Received **CDN logs** containing the selected bitrates and QoE parameters of the players
- Be **end-to-end** and **agnostic** of the player/ABR types
- Operate in a **time-slotted fashion**

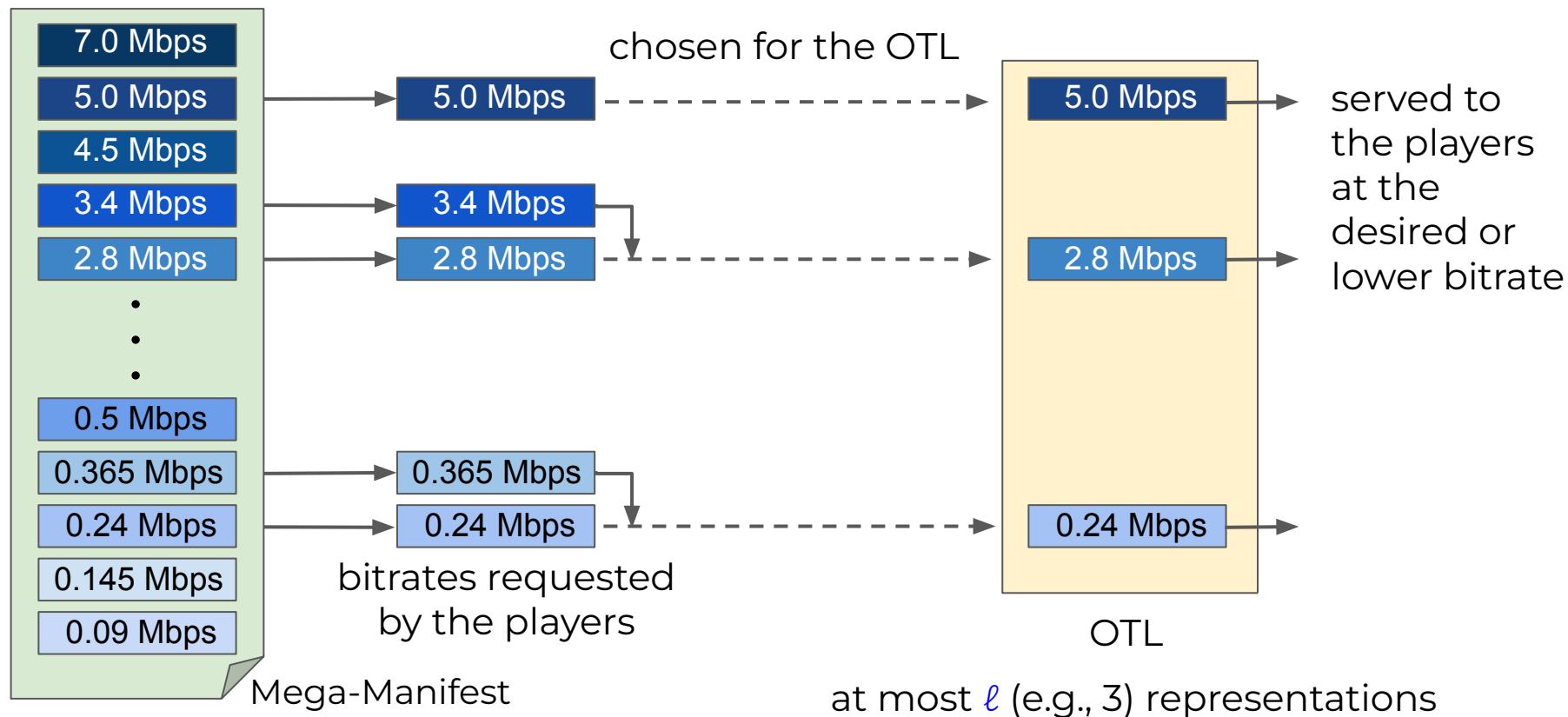
ARTEMIS at a Glance



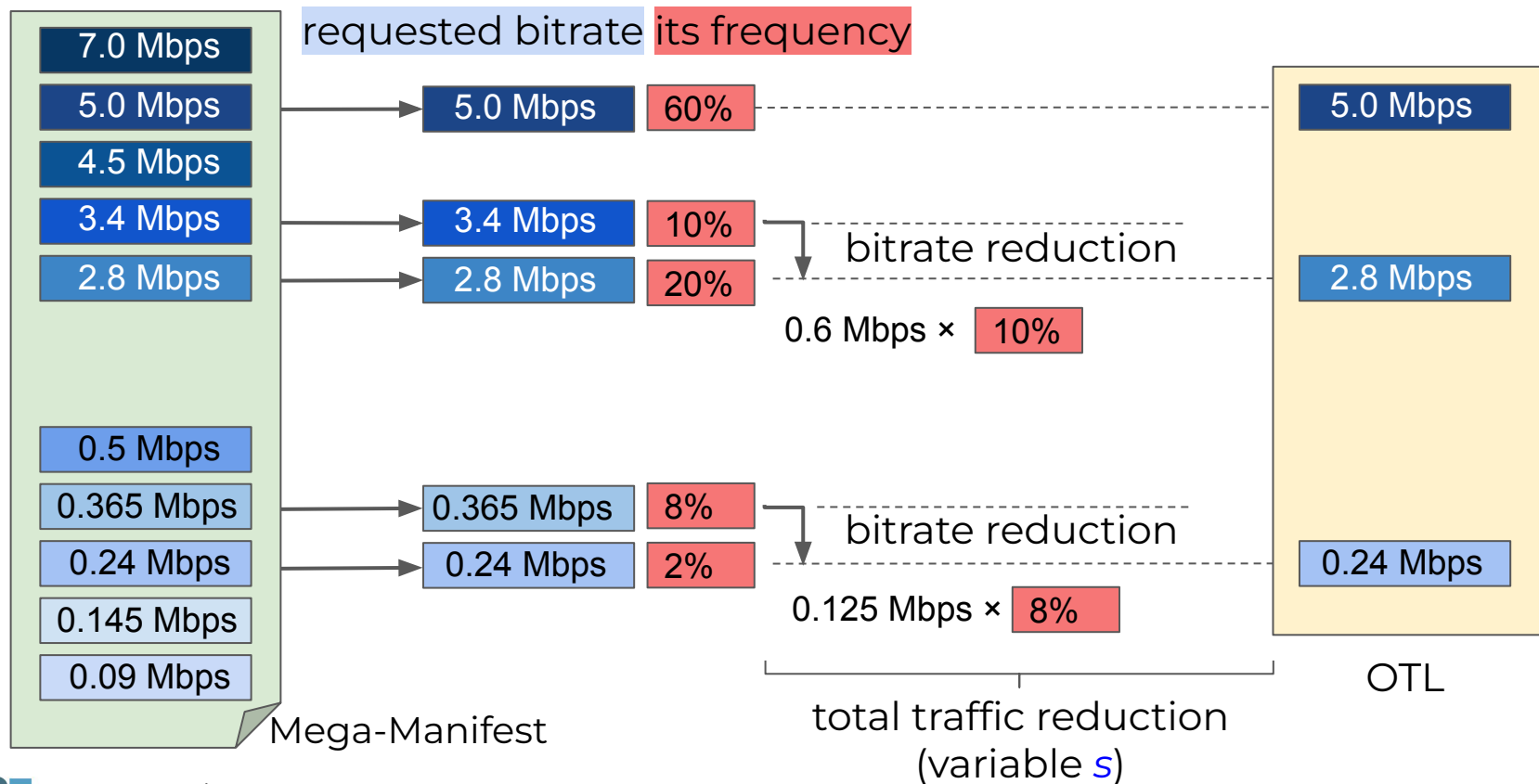
Optimal Temporary Ladder (OTL)

- Cornerstone of ARTEMIS
- **Optimal subset of the mega-manifest** representations utilized by the live encoder
- **Challenges**
 - How to select the OTL?
 - Which constraints should the OTL impose?
 - When should the OTL be updated?

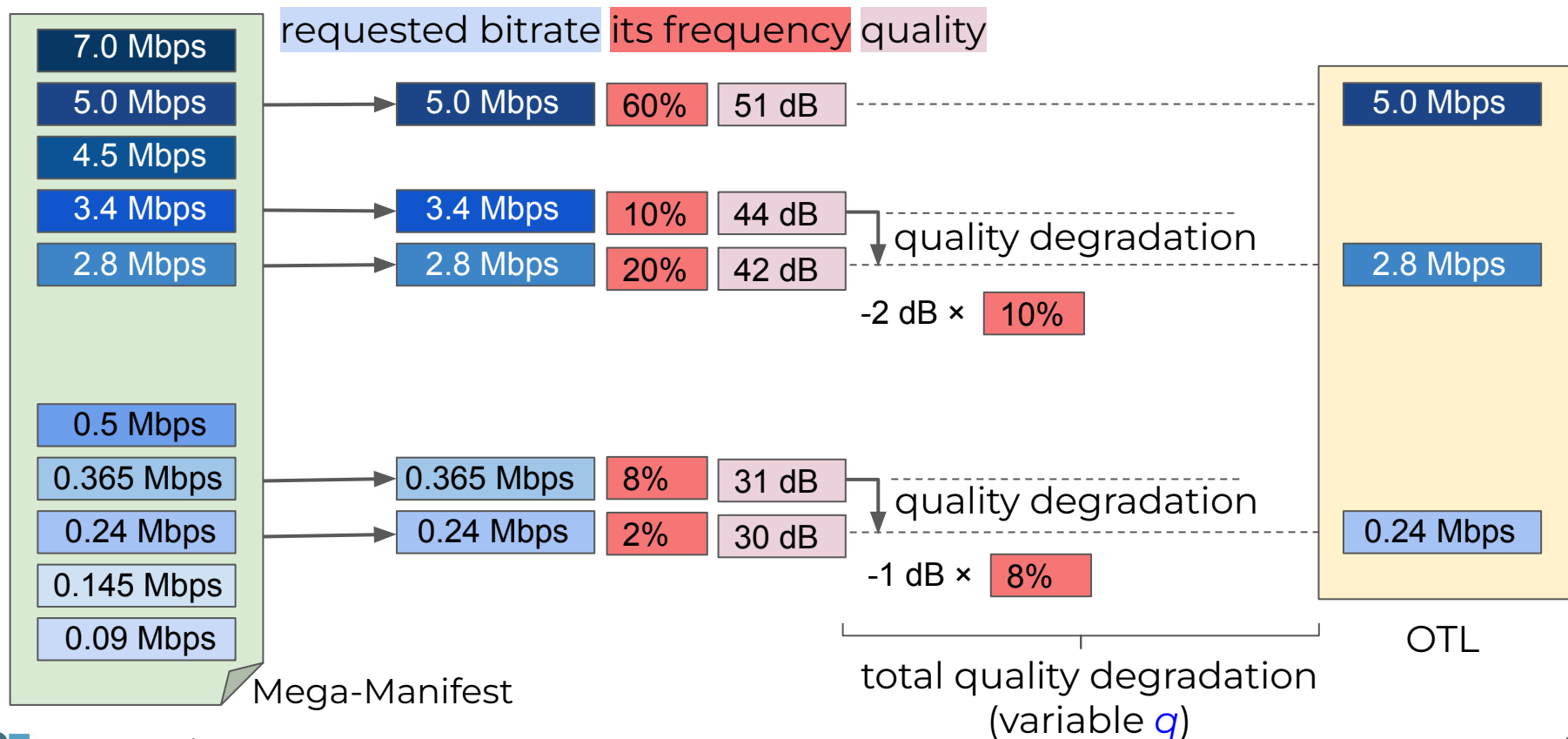
OTL : Maximum Length and Bitrate Reduction



OTL : Traffic Reduction



OTL : Quality Improvement



OTL : Computation

- Objective function:

$$\text{Maximize } (1 - \alpha) \times s - \alpha \times q$$

- Parameter α

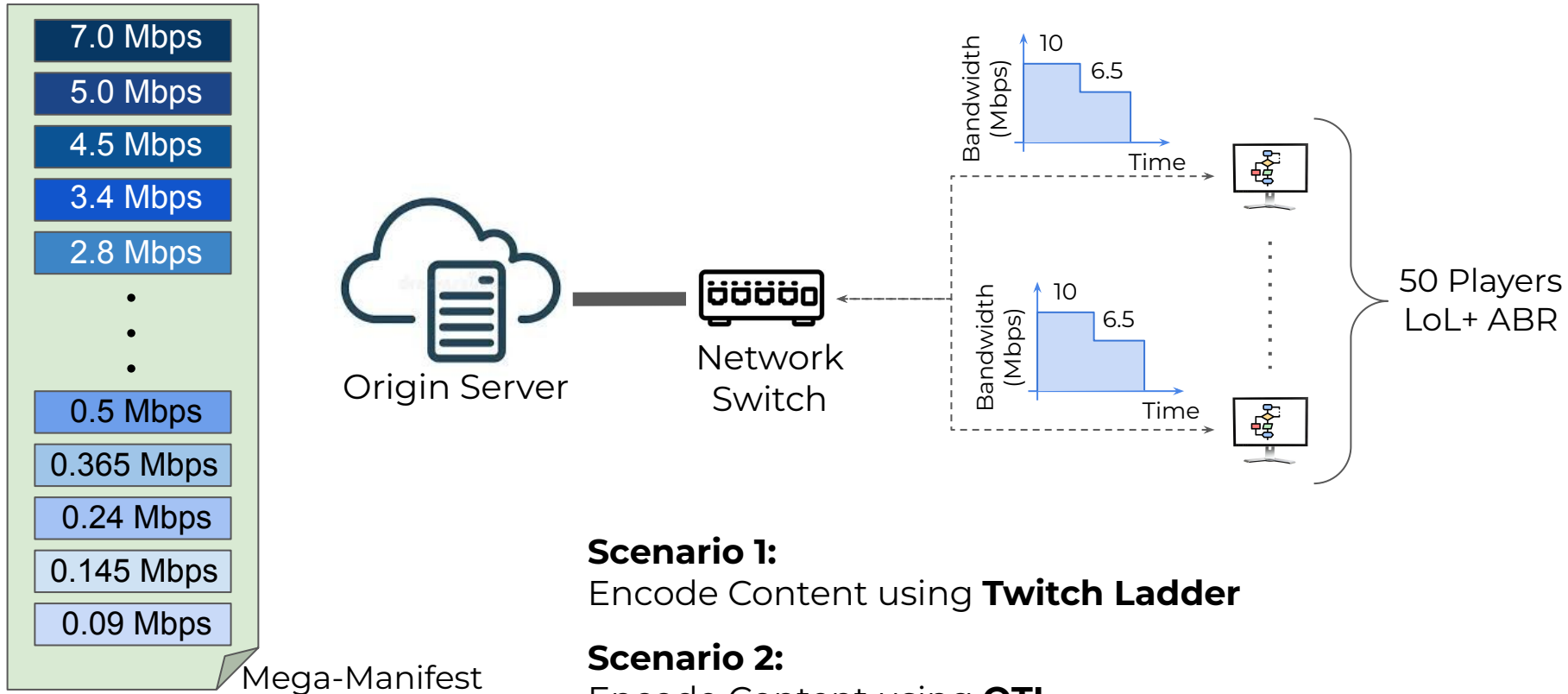
- Relative importance of quality improvement vs. traffic reduction
- Computed from the stall information of the players

Evaluation Settings

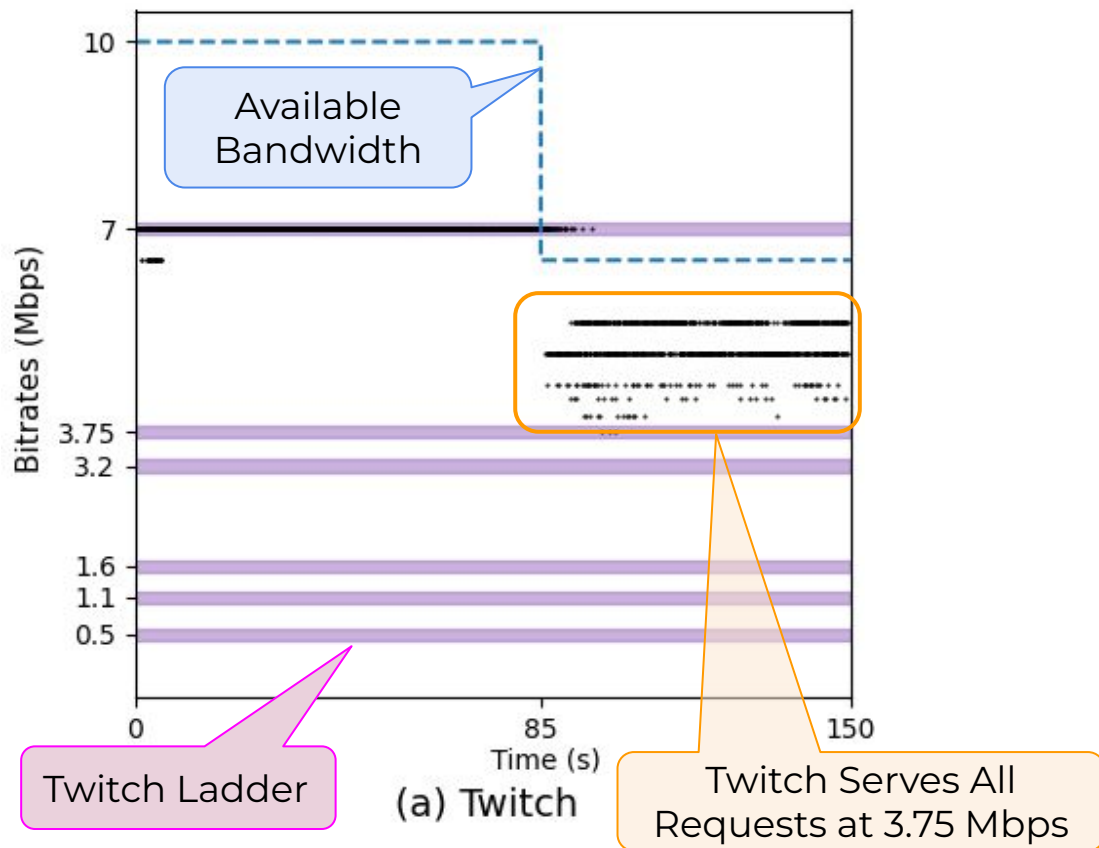
- Network traces: LTE, AmazonFCC, Cascade-5, and Cascade-20
- Content type: animation, sport, movie, and documentary
- Baselines for ARTEMIS: five static bitrate ladders
- Players: dash.js

Bitrate Ladder	Length	Min. [Res.@Bitrate]	Max. [Res.@Bitrate]
Theo	4	360p@0.365 Mbps	1080p@4.0 Mbps
Mux	4	360p@0.75 Mbps	1080p@4.5 Mbps
Bitmovin	6	240p@0.145 Mbps	1080p@4.5 Mbps
Pensieve	6	360p@0.365 Mbps	1080p@4.3 Mbps
Twitch	6	360p@0.5 Mbps	1080p@7.0 Mbps
ARTEMIS	29	240p@0.145 Mbps	1080p@7.0 Mbps

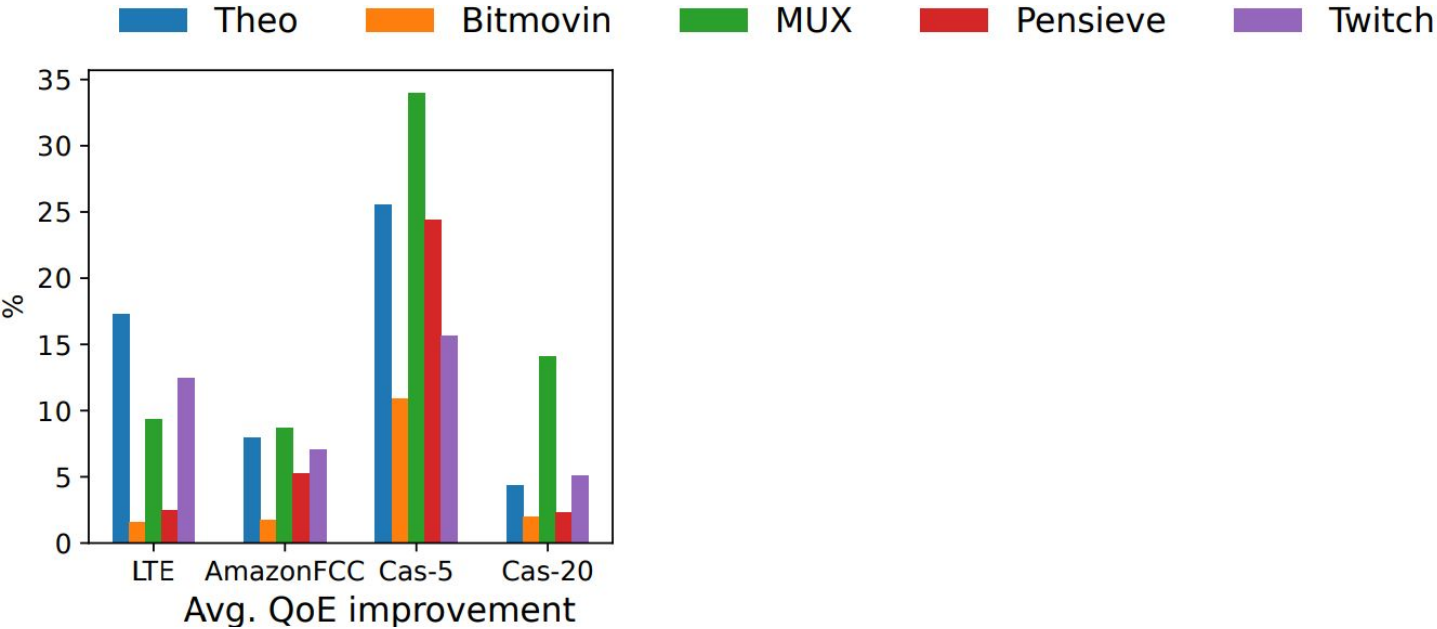
Static Ladder (Twitch) vs. OTL : Experimental Setup



Static Ladder (Twitch) vs. OTL

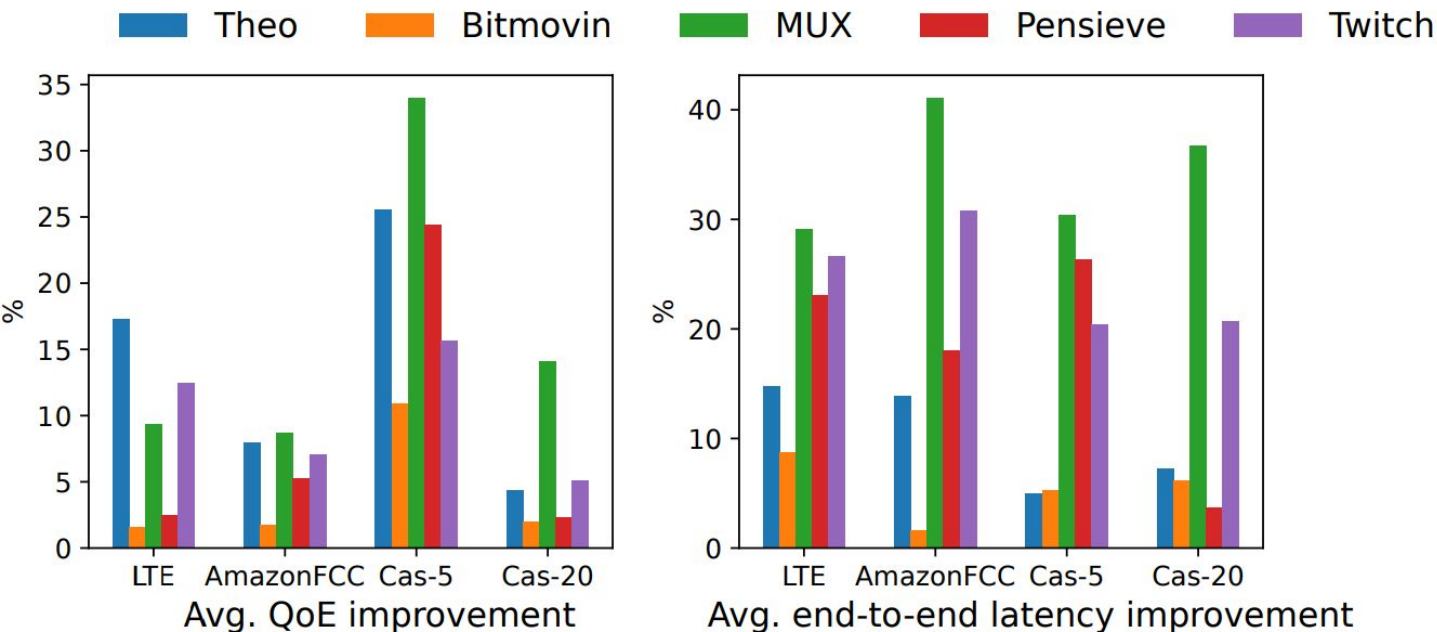


Quality of Experience (QoE) with ARTEMIS vs. Five Static Bitrate Ladders



Average QoE up 11%

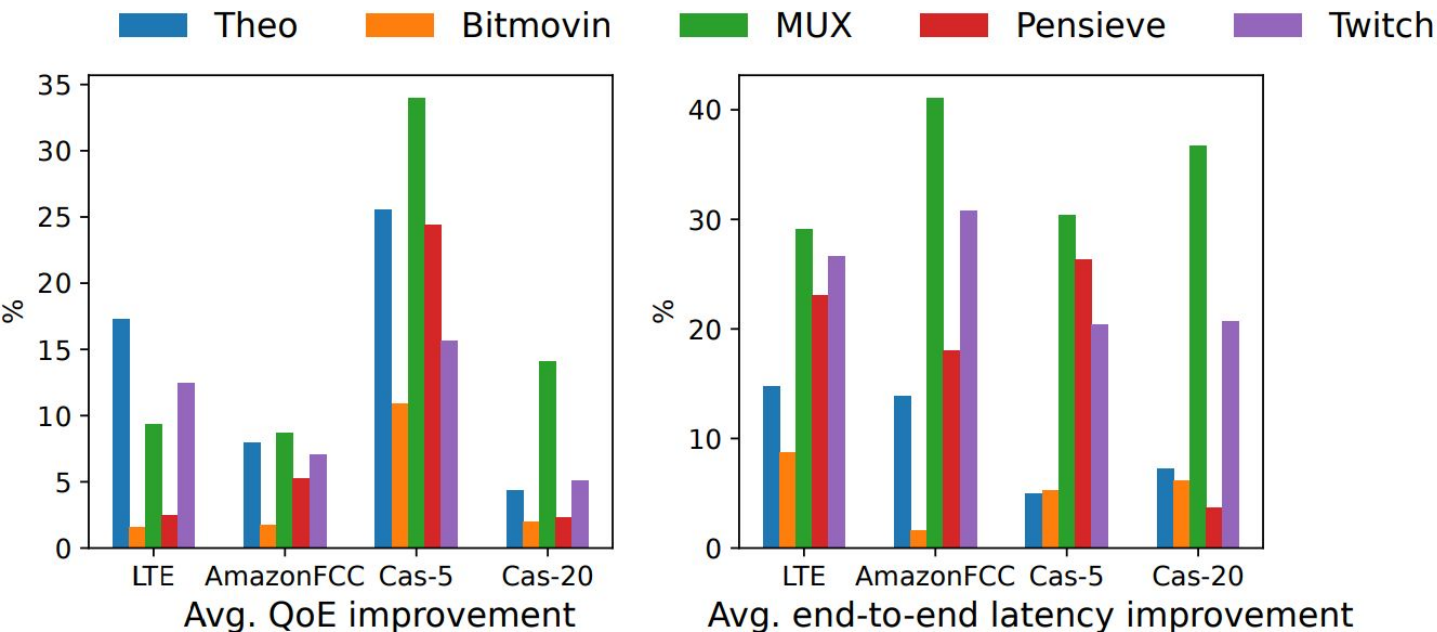
QoE and **Latency** with ARTEMIS vs. Five Static Bitrate Ladders



Average QoE up 11%

Average Latency down 18%

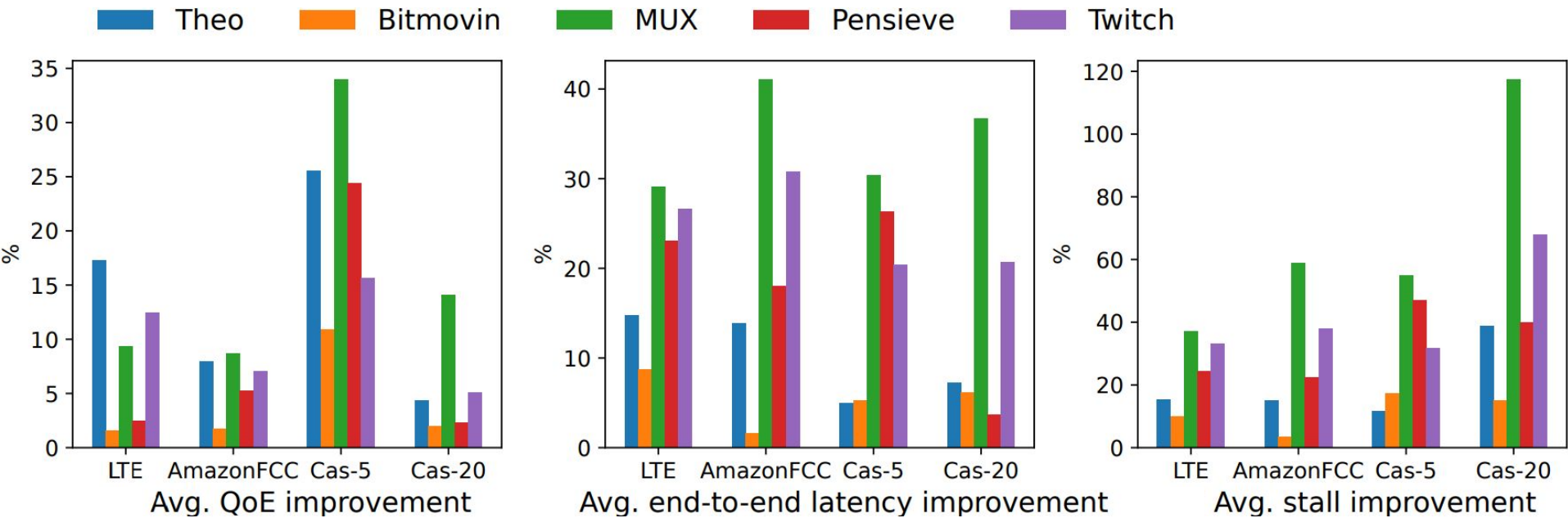
QoE and **Latency** with ARTEMIS vs. Five Static Bitrate Ladders



Average QoE up 11%

Average Latency down 18%

QoE, Latency, and **Stall** with ARTEMIS vs. Five Static Bitrate Ladders



Average QoE up 11%

Average Latency down 18%

Average Stall down 36%

ARTEMIS : Summary

- **Adaptive bitrate ladder optimization** for live video streaming
 - Seamless enhancement of the **end-to-end** pipeline
 - Accounting for the **context** via scalable player feedback
 - **Content** awareness via PSNR
- **Mega-manifest** to advertise a large number of representations
- **Short dynamic ladder for encoding** the content
 - An optimal subset of the mega-manifest representations
- Multi-objective **performance improvement**
 - Reduced **end-to-end** latency and **stall**
 - Increased **quality of experience**

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